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United States Patent	12397335
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Onagi; Yousuke

Bending system

Abstract

On a back side of a table, moving bodies are provided being movable in a left-right direction. A bending system includes moving actuators to cause the moving bodies to move in the left-right direction relative to the table and position detectors to detect positions of the moving bodies in the left-right direction. The moving bodies are provided with hole detectors to detect the engagement holes of tools fitted to a tool holder. The bending system includes a hole position acquisition unit to acquire positions of the engagement holes of the tools in the left-right direction, based on positions of the moving bodies in the left-right direction at the time of the engagement holes of the tools being detected.

Inventors:	Onagi; Yousuke (Kanagawa, JP)
Applicant:	AMADA CO., LTD. (Kanagawa, JP)
Family ID:	1000008780342
Assignee:	AMADA CO., LTD. (Kanagawa, JP)
Appl. No.:	17/615616
Filed (or PCT Filed):	June 03, 2020
PCT No.:	PCT/JP2020/021960
PCT Pub. No.:	WO2020/246507
PCT Pub. Date:	December 10, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220234088 A1	Jul. 28, 2022

Foreign Application Priority Data

JP	2019-107126	Jun. 07, 2019
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Publication Classification

Int. Cl.: B21D5/02 (20060101); B21D5/00 (20060101); B21D37/04 (20060101); B21D37/14 (20060101)

U.S. Cl.:

CPC B21D5/004 (20130101); B21D5/0254 (20130101); B21D37/04 (20130101); B21D37/145 (20130101); Y10T483/136 (20150115); Y10T483/1731 (20150115)

Field of Classification Search

CPC: B21D (5/02); B21D (5/0254); B21D (37/04); B21D (37/145); B21D (5/004); Y10T (483/13-138); Y10T (483/1729); Y10T (483/1731)

USPC: 483/28; 483/29; 483/7-11

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Primary Examiner: Singh; Sunil K

Assistant Examiner: Vitale; Michael

Attorney, Agent or Firm: Renner, Otto, Boisselle & Sklar, LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a bending system for performing bending for a plate-shaped workpiece (a sheet metal).

BACKGROUND ART

(2) A bending system includes a press brake to perform bending for a workpiece, and the press brake includes upper and lower tables and tool holders provided on the respective tables and configured to retain tools. Further, the bending system includes a tool rack arranged on the side of the press brake and configured to store a plurality of tools. The tool rack has a plurality of stockers for retaining tools in which selected arbitrary stockers are configured such that the stockers can be positioned at exchange positions for performing tool exchange. Furthermore, the bending system includes tool exchange units for transporting tools between the tool holders and stockers positioned at the exchange positions, and each tool exchange unit has a tool retention member that is engageable with and disengageable from an engagement hole of a tool (see Patent Literature 1).

(3) As prior-art techniques related to the present disclosure, techniques shown in Patent Literatures 2 to 4 exist in addition to the technique of Patent Literature 1.

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Patent No. 5947861 Patent Literature 2: Japanese Patent No. 4672868 Patent Literature 3: Japanese Patent No. 4558852 Patent Literature 4: Japanese Patent Application Laid-Open Publication No. 2003-71519

SUMMARY

(5) A conventional bending system does not recognize actual positions of tools in a left-right direction on tool holders. Therefore, the conventional bending system has the following problem.

(6) For example, if a tool is fitted to a position deviated from a regular arrangement position on a tool holder corresponding to tool layout information, and engagement or disengagement of a tool retention member with an engagement hole of a tool (a retention operation) fails, an alarm occurs, and an operation related to tool transportation by a tool exchange unit stops. Further, for example, if a clamp plate or a clamper of a tool holder is accidentally released after tools are fitted to the tool holder, there is a possibility that the tools slightly move and are displaced from regular arrangement

positions on the tool holder, and, therefore, it is not possible to perform the operation related to tool transportation by a tool exchange unit. Furthermore, for example, tools are randomly laid out on a tool holder by manual work or the like, it is not possible to lay out again the tools at the regular arrangement positions on the tool holder by a tool exchange unit. That is, in the former case, there is a problem that it becomes difficult to stabilize automatic operation of the bending system. Here, the operation related to tool transportation is assumed to include an operation of exchanging tools fitted to a tool holder with tools fitted to a stocker, an operation of retaining a tool by a tool retention member and an operation of laying out tools on a tool holder or a stocker.

(7) In a case where tools are randomly laid out on a tool holder by manual work or the like in a general-purpose bending system without tool exchange units, problems similar to the above also occur.

(8) Therefore, an object of the present disclosure is to provide a bending system capable of accurately recognizing actual positions of tools in a left-right direction on a tool holder or on a stocker.

(9) A bending system according to a first aspect of one or more embodiments includes: a moving body provided on a back side or a front side of a table of a press brake so as to be movable in a left-right direction; a moving actuator configured to cause the moving body to move in the left-right direction relative to the table; and a position detector configured to detect a position of the moving body in the left-right direction. The bending system according to a first embodiment includes: a hole detector provided on the moving body and configured to detect an engagement hole for causing a tool retention member of a tool exchange unit to be engaged and disengaged, the engagement hole being a part of a tool fitted to a tool holder provided on the table; and a hole position acquisition unit configured to acquire a position of the engagement hole of the tool in the left-right direction, based on the position of the moving body in the left-right direction at the time of the engagement hole of the tool being detected.

(10) The bending system according to the first aspect may further include an information reader provided on the moving body and configured to read, from a storage medium provided to the tool, tool information including a width dimension of the tool in the left-right direction and a relative position of the engagement hole in the tool.

(11) According to the first aspect, the moving body is moved in the left-right direction relative to the table by driving of the moving actuator while the position of the moving body in the left-right direction is being detected by the position detector. Further, during the movement of the moving body, the engagement hole of the tool is detected by the tool detector. Then, the hole position acquisition unit acquires the position of the engagement hole of the tool in the left-right direction, based on the position of the moving body in the left-right direction at the time of the engagement hole of the tool being detected. Thereby, the bending system can recognize an actual position of the engagement hole of the tool in the left-right direction on the tool holder.

(12) A bending system according to a second aspect of one or more embodiments further includes: a moving body provided on a back side or a front side of a table of a press brake so as to be movable in a left-right direction; a moving actuator configured to cause the moving body to move in the left-right direction relative to the table; and a position detector configured to detect a position of the moving body in the left-right direction. The bending system according to the second aspect includes: an end face detector provided on the moving body and configured to detect an end face of a tool fitted to a tool holder provided on the table; and an information reader provided on the moving body and configured to read, from a storage medium provided to the tool, tool information including a width dimension of the tool in the left-right direction and a relative position of an engagement hole in the tool, the engagement hole being a part of the tool and being for causing a tool retention member of a tool exchange unit to be engaged and disengaged. The bending system according to the second aspect further includes: a hole position acquisition unit configured to acquire a position of the engagement hole of the tool in the left-right direction, based on the

position of the moving body in the left-right direction at the time of the end face of the tool being detected, and the read tool information about the tool.

(13) According to the second aspect, the moving body is moved in the left-right direction by driving of the moving actuator while the position of the moving body in the left-right direction is being detected by the position detector. During the movement of the moving body, the end face of the tool is detected by the tool detector, and the tool information about the tool is read by the information reader. Then, the hole position acquisition unit acquires the position of the engagement hole of the tool in the left-right direction, based on the position of the moving body in the left-right direction at the time of the end face of the tool being detected, and the read tool information about the tool. Thereby, the bending system can recognize an actual position of the engagement hole of the tool in the left-right direction on the tool holder.

(14) A bending system according to a third aspect of one or more embodiments includes: a moving body provided on a back side or a front side of a table of a press brake so as to be movable in a left-right direction; a moving actuator configured to cause the moving body to move in the left-right direction relative to the table; and a position detector configured to detect a position of the moving body in the left-right direction. The bending system according to the third aspect further includes: an end face detector provided on the moving body and configured to detect an end face of a tool fitted to a tool holder provided on the table; and an information reader provided on the moving body and configured to read, from a storage medium provided to the tool, tool information including a width dimension of the tool in the left-right direction. The bending system according to the third aspect further includes a tool position acquisition unit configured to acquire a position of the tool in the left-right direction (a position of the end face of the tool in the left-right direction or a position of the center of the tool in the left-right direction), based on the position of the moving body in the left-right direction at the time of the end face of the tool being detected, and the read tool information about the tool.

(15) According to the third aspect, the moving body is moved in the left-right direction by driving of the moving actuator while the position of the moving body in the left-right direction is being detected by the position detector. During the movement of the moving body, the engagement hole of the tool is detected by the hole detector, and the tool information about the tool is read by the information reader. Then, the tool position acquisition unit acquires the position of the tool in the left-right direction, based on the position of the moving body in the left-right direction at the time of the engagement hole of the tool being detected, and the read tool information about the tool. Thereby, the bending system can recognize an actual position of the tool in the left-right direction on the tool holder.

(16) According to the present disclosure, it is possible to accurately recognize an actual positions of a tool in a left-right direction on a tool holder or stocker.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic front view of a bending system according to first and second embodiments.

(2) FIG. 2 is a schematic right side view of the bending system according to the first embodiment, and shows that die tools are retained by a lower tool retention member, and punch tools are retained by an upper tool retention member. In FIG. 2, a lower joint block and an upper joint block are omitted.

(3) FIG. 3 is an enlarged diagram of a part III in FIG. 2.

(4) FIG. 4(a) is a schematic diagram showing that an engagement hole of a die tool is being detected, and that identification information about the die tool is being read in the first

embodiment. FIG. 4(b) is a schematic diagram showing that an inspection beam moves in a left-right direction in the first embodiment.

(5) FIG. 5(a) is a schematic diagram showing that an engagement hole of a punch tool is being detected, and that identification information about the punch tool is being read in the first embodiment. FIG. 5(b) is a schematic diagram showing that an inspection beam moves in the left-right direction in the first embodiment.

(6) FIG. 6 is a control block diagram of the bending system according to the first and second embodiments.

(7) FIGS. 7A(a) and 7A(b) are schematic diagrams illustrating a retry process of a lower tool exchange unit and the like according to the first embodiment.

(8) FIGS. 7B(a) and 7B(b) are schematic diagrams illustrating a retry process of the lower tool exchange unit and the like according to the first embodiment.

(9) FIGS. 8A(a) and 8A(b) are schematic diagrams illustrating a layout process of the lower tool exchange unit and the like.

(10) FIGS. 8B(a) and 8B(b) are schematic diagrams illustrating a layout process of the lower tool exchange unit and the like.

(11) FIG. 9(a) is a schematic diagram showing that an end face of a die tool is being detected, and that identification information about the die tool is being read in a second embodiment. FIG. 9(b) is a schematic diagram showing that an inspection beam moves in the left-right direction in the second embodiment.

(12) FIG. 10(a) is a schematic diagram showing that an end face of a punch tool is being detected, and that identification information about the punch tool is being read in the second embodiment. FIG. 10(b) is a schematic diagram showing that the inspection beam moves in the left-right direction in the second embodiment.

(13) FIGS. 11A(a) and 11A(b) are schematic diagrams illustrating a retry process of a lower tool exchange unit and the like according to the second embodiment.

(14) FIGS. 11B(a) and 11B(b) are schematic diagrams illustrating a retry process of the lower tool exchange unit and the like according to the second embodiment.

(15) FIGS. 12A(a) and 12A(b) are schematic diagrams illustrating a layout process of the lower tool exchange unit and the like according to the second embodiment.

(16) FIGS. 12B(a) and 12B(b) are schematic diagrams illustrating a layout process of the lower tool exchange unit and the like according to the second embodiment.

(17) FIG. 13 is a control block diagram of a bending system according to a third embodiment.

(18) FIGS. 14(a) and 14(b) are schematic diagrams illustrating a process of detecting an end face of a die tool in the third embodiment.

(19) FIG. 15 is a control block diagram of a bending system according to a fourth embodiment.

DESCRIPTION OF EMBODIMENTS

(20) First to fourth embodiments will be explained below with reference to FIGS. 1 to 15.

(21) In the specification and Claims of the present application, “being provided” is assumed to include being indirectly provided via another member in addition to being directly provided. “A left-right direction” is one of horizontal directions and is synonymous with a lateral direction of a press brake, a tool rack, a tool holder or a tool. “A front-back direction” refers to a horizontal direction orthogonal to the left-right direction and is synonymous with a depth direction of the press brake or the tool rack. “Read tool information about a tool” is assumed to include reading identification information about a tool corresponding to tool information about the tool. In the drawings, “FF”, “FR”, “L”, “R”, “U” and “D” refer to a front direction, a back direction, a left direction, a right direction, an up direction and a down direction, respectively.

First Embodiment

(22) As shown in FIGS. 1 to 3, a bending system 10 according to a first embodiment is a system for performing bending for a plate-shaped workpiece (a sheet metal) W using punch tools 12 as

automatically exchangeable upper tools and die tools **14** as lower tools. Further, the bending system **10** includes a press brake **16** to perform bending for the workpiece **W** in cooperation with the punch tools **12** and the die tools **14**. First, a configuration of the press brake **16** will be specifically explained.

(23) The press brake **16** includes a main frame **18**, and the main frame **18** has a pair of side plates **20** separated in the left-right direction and facing each other, and a plurality of coupling members **22** to couple the pair of side plates **20**. Further, at a lower part of the main frame **18**, a lower table **24** extending in the left-right direction is provided. At an upper part of the main frame **18**, an upper table **26** extending in the left-right direction is provided being movable in a vertical direction. At an upper part of each side plate **20**, a hydraulic cylinder **28** is provided as a vertically moving actuator to cause the upper table **26** to move in the vertical direction relative to the main frame **18**.

(24) The lower table **24** may be configured to be movable in the vertical direction instead of configuring the upper table **26** to be movable in the vertical direction. As vertically moving actuators, servo motors (not shown) may be used instead of the hydraulic cylinders **28**.

(25) On the upper side of the lower table **24**, a lower tool holder **30** to removably retain die tools **14** is provided, and the lower tool holder **30** extends in the left-right direction. On the lower tool holder **30**, a holder groove **30g** into which shanks (bases) **14s** of the die tools **14** are to be inserted is formed along the left-right direction, the holder groove **30g** being in a publicly known configuration as shown in Patent Literatures 2 and 3. The lower tool holder **30** has a hydraulic clamber **32** to fix the die tools **14** to the lower table **24**.

(26) Here, as shown in FIG. 4, in the center of each die tool **14** in the left-right direction (the lateral direction), an engagement hole **14h** as a part of the die tool **14** is penetratingly formed. At positions corresponding to the engagement hole **14h** (for example, on the lower side of the engagement hole **14h**) on both faces (the front face and back face) of the die tool **14**, identification marks **14m** as storage media storing identification information about the die tool **14** are provided by engraving, and each identification mark **14m** is configured with a two-dimensional code. The identification information about the die tool **14** corresponds to tool information about the die tool **14**, and the tool information about the die tool **14** includes a type, shape, width dimension in the left-right direction and height dimension of the die tool **14**, a relative position of the center of the engagement hole **14h** in the die tool **14** and the like. The width dimension of the die tool **14** in the left-right direction is assumed to include a dimension from an end face **14e** of the die tool **14** to the center of the die tool **14** in the left-right direction. The relative position of the center of the engagement hole **14h** in the die tool **14** is assumed to include a dimension from the end face **14e** of the die tool **14** to the center of the engagement hole **14h** in the left-right direction and an amount of deviation of the center of the engagement hole **14h** relative to the center of the die tool **14**.

(27) As shown in FIGS. 1 to 3, an upper tool holder **34** to removably retain the punch tools **12** is provided on the lower side of the upper table **26**, and the upper tool holder **34** extends in the left-right direction. On the upper tool holder **34**, a holder groove **34g** into which shanks (bases) **12s** of punch tools **12** are to be inserted is formed along the left-right direction, the holder groove **34g** being in a publicly known configuration as shown in Patent Literatures 2 and 3. The upper tool holder **34** has a hydraulic clamp plate **36** to fix the punch tools **12** to the upper table **26**.

(28) Here, as shown in FIG. 5, in the center of each punch tool **12** in the left-right direction, an engagement hole **12h** as a part of the punch tool **12** is penetratingly formed. At positions corresponding to the engagement hole **12h** (for example, on the upper side of the engagement hole **12h**) on both faces of the punch tool **12**, identification marks **12m** as storage media storing identification information about the punch tool **12** are provided by engraving, and each identification mark **12m** is configured with a two-dimensional code. The identification information about the punch tool **12** corresponds to tool information about the punch tool **12**, and the tool information about the punch tool **12** includes a type, shape, width dimension in the left-right direction and height dimension of the punch tool **12**, a relative position of the center of the

engagement hole **12h** in the punch tool **12** and the like. The width dimension of the punch tool **12** in the left-right direction is assumed to include a dimension from an end face **12e** of the punch tool **12** to the center in the left-right direction. The relative position of the center of the engagement hole **12h** in the punch tool **12** is assumed to include a dimension from the end face **12e** of the punch tool **12** to the center of the engagement hole **12h** in the left-right direction and an amount of deviation of the center of the engagement hole **12h** relative to the center of the punch tool **12**.

(29) As shown in FIG. 1, on one side (the right side) of the lower tool holder **30** in the left-right direction, a lower joint block **38** extending in the left-right direction is provided. On the lower joint block **38**, a joint groove (not shown) into which the shank **14s** of a die tool **14** is to be inserted is formed along the left-right direction. The joint groove of the lower joint block **38** is connected to the holder groove **30g** (see FIG. 3) of the lower tool holder **30**. Further, on one side of the upper tool holder **34** in the left-right direction, an upper joint block **40** extending in the left-right direction is provided. On the upper joint block **40**, a joint groove (not shown) into which the shank **12s** of a punch tool **12** are to be inserted is formed along the left-right direction. The joint groove of the upper joint block **40** is connected to the holder groove **34g** (see FIG. 3) of the upper tool holder **34**.

(30) As shown in FIGS. 1 and 2, the bending system **10** includes a tool rack **42** arranged on the side (the right side) of the press brake in the left-right direction and storing a plurality of die tools **14** and a plurality of punch tools **12**. The tool rack **42** is, for example, in a publicly known configuration as shown in Patent Literatures 1 and 4. The configuration of the tool rack **42** will be simply explained.

(31) The tool rack **42** has a plurality of lower stockers **44** for retaining one or more die tools **14**, and the plurality of lower stockers **44** are arranged in the front-back direction. Each lower stocker **44** extends in the left-right direction, and, on each lower stocker **44**, a stocker groove **44g** into which the shanks **14s** of die tools **14** are to be inserted is formed along the left-right direction.

(32) A configuration is made so that a selected arbitrary lower stocker **44** is movable in the front-back direction by such a lower stocker movement mechanism (not shown) as shown in Patent Literature 1. Alternatively, the plurality of lower stockers **44** are configured to be synchronously movable in the front-back direction by such a lower stocker movement mechanism (not shown) as shown in Patent Literature 4. A configuration is made so that the selected arbitrary lower stocker **44** is positioned at a lower tool exchange position contiguous to the lower joint block **38** in the left-right direction, by its movement in the front-back direction. When the arbitrary lower stocker **44** is positioned at the lower tool exchange position, the stocker groove **44g** of the lower stocker **44** is connected to the joint groove of the lower joint block **38**.

(33) The tool rack **42** has a plurality of upper stockers **46** for retaining one or more punch tools **12**, and the plurality of upper stockers **46** are arranged along the front-back direction. Each upper stocker **46** extends in the left-right direction, and, on each upper stocker **46**, a stocker groove **46g** into which the shanks **12s** of punch tools **12** are to be inserted is formed along the left-right direction.

(34) A configuration is made so that a selected arbitrary upper stocker **46** is movable in the front-back direction by such an upper stocker movement mechanism (not shown) as shown in Patent Literature 1. Alternatively, the plurality of upper stockers **46** are configured to be synchronously movable in the front-back direction by such an upper stocker movement mechanism (not shown) as shown in Patent Literature 4. A configuration is made so that the selected arbitrary upper stocker **46** is positioned at an upper tool exchange position contiguous to the upper joint block **40** in the left-right direction, by its movement in the front-back direction. When the arbitrary upper stocker **46** is positioned at the upper tool exchange position, the stocker groove **46g** of the upper stocker **46** is connected to the joint groove of the upper joint block **40**.

(35) As shown in FIGS. 1 to 3, a lower guide **48** extending in the left-right direction is provided on the back side (the rear side) of the lower tool holder **30**, and the lower guide **48** projects to the right of the lower joint block **38**. On the back side of the lower table **24**, a pair of lower tool exchange

units **50** as lower moving bodies are provided being movable in the left-right direction via the lower guide **48**. The lower tool exchange units **50** transport die tools **14** between the lower tool holder **30** and a lower stocker **44** positioned at the lower tool exchange position. Each lower tool exchange unit **50** moves in the left-right direction relative to the lower table **24** by driving of a servo motor **52** as a left-right moving actuator provided at an appropriate position on the lower tool exchange unit **50**. The servo motor **52** has an encoder **54** as a position detector to detect a position of the lower tool exchange unit **50** in the left-right direction.

(36) Each lower tool exchange unit **50** is in a publicly known configuration as shown in Patent Literatures 2 and 3, and has a lower unit body **56** provided on the lower guide **48** being movable in the left-right direction and a lower support member **58** provided on the lower unit body **56** being movable in the front-back direction and in the vertical direction. The lower support member **58** moves in the front-back direction relative to the lower unit body **56** by driving of an air cylinder **60** as a front-back moving actuator provided at an appropriate position on the lower unit body **56**. The lower support member **58** moves in the vertical direction relative to the lower unit body **56** by driving of an air cylinder **62** as a vertically moving actuator provided at an appropriate position on the lower unit body **56**.

(37) Each lower tool exchange unit **50** has a lower tool retention member (a lower hook member) **64** provided on the lower support member **58** being movable in the front-back direction, and the lower tool retention member **64** is engageable with and disengageable from the engagement hole **14h** of a die tool **14**. The lower tool retention member **64** moves in the front-back direction relative to the lower support member **58** by driving of an air cylinder **66** as another front-back moving actuator provided at an appropriate position on the lower support member **58**.

(38) Since an operation of retaining a die tool **14** by the lower tool retention member **64** is commonly known as shown in Patent Literatures 2 and 3, details thereof will be omitted.

(39) As shown in FIGS. **3** and **4**, at an appropriate position on each lower support member **58**, a photoelectric sensor **68** is provided as a hole detector to detect the engagement hole **14h** of a die tool **14** fitted to the lower tool holder **30**. The photoelectric sensor **68** is configured to emit an inspection beam B and detect the engagement hole **14h** of the die tool **14** from a result of receiving a reflected beam of the inspection beam B, in a state of being located at the same height as the engagement hole **14h** of the die tool **14**.

(40) On the lower side of the photoelectric sensor **68** on each lower support member **58**, a two-dimensional code reader **70** as an information reader is provided, and the two-dimensional code reader **70** reads tool identification information about a die tool **14** from an identification mark **14m** in a state of being located at the same height as the identification mark **14m**. In other words, on the lower side of the photoelectric sensor **68** on the lower support member **58**, the two-dimensional code reader **70** to read tool information about the die tool **14** via the tool identification information about the die tool **14** is provided.

(41) The photoelectric sensor **68** and the two-dimensional code reader **70** may be provided on the lower unit body **56** instead of being provided on the lower support member **58**.

(42) As shown in FIGS. **1** to **3**, an upper guide **72** extending in the left-right direction is provided on the back side of the upper tool holder **34**, and the upper guide **72** projects to the right of the upper joint block **40**. On the back side of the upper table **26**, a pair of upper tool exchange units **74** as upper moving bodies are provided being movable in the left-right direction via the upper guide **72**. The upper tool exchange units **74** transport punch tools **12** between the upper tool holder **34** and an upper stocker **46** positioned at the upper tool exchange position. Each upper tool exchange unit **74** moves in the left-right direction relative to the upper table **26** by driving of a servo motor **76** as a left-right moving actuator provided at an appropriate position on the upper tool exchange unit **74**. The servo motor **76** has an encoder **78** as a position detector to detect a position of the upper tool exchange unit **74** in the left-right direction.

(43) Each upper tool exchange unit **74** is in a publicly known configuration as shown in Patent

Literatures 2 and 3, and has an upper unit body **80** provided on the upper guide **72** being movable in the left-right direction and an upper support member **82** provided on the upper unit body **80** being movable in the front-back direction and in the vertical direction. The upper support member **82** moves in the front-back direction relative to the upper unit body **80** by driving of an air cylinder **84** as a front-back moving actuator provided at an appropriate position on the upper unit body **80**. The upper support member **82** moves in the vertical direction relative to the upper unit body **80** by driving of an air cylinder **86** as a vertically moving actuator provided at an appropriate position on the upper unit body **80**.

(44) Each upper tool exchange unit **74** has an upper tool retention member (an upper hook member) **88** provided on the upper support member **82** being movable in the front-back direction, and the upper tool retention member **88** is engageable with and disengageable from the engagement hole **12h** of a punch tool **12**. The upper tool retention member **88** moves in the front-back direction relative to the upper support member **82** by driving of an air cylinder **90** as another front-back moving actuator provided at an appropriate position on the upper support member **82**.

(45) Since an operation of retaining a punch tool **12** by the upper tool retention member **88** is commonly known as shown in Patent Literatures 2 and 3, details thereof will be omitted.

(46) As shown in FIGS. **3** and **5**, at an appropriate position on each upper support member **82**, a photoelectric sensor **92** is provided as a hole detector to detect the engagement hole **12h** of a punch tool **12** fitted to the upper tool holder **34**. The photoelectric sensor **92** is configured to emit an inspection beam B and detect the engagement hole **12h** of the punch tool **12** from a result of receiving a reflected beam of the inspection beam B, in a state of being located at the same height as the engagement hole **12h** of the punch tool **12**.

(47) On the upper side of the photoelectric sensor **92** on each upper support member **82**, a two-dimensional code reader **94** as an information reader is provided, and the two-dimensional code reader **94** reads identification information about a punch tool **12** from an identification mark **12m** in a state of being located at the same height as the identification mark **12m**. In other words, on the upper side of the photoelectric sensor **92** on the upper support member **82**, the two-dimensional code reader **94** to read tool information about the punch tool **12** via the identification information about the punch tool **12** is provided.

(48) The photoelectric sensor **92** and the two-dimensional code reader **94** may be provided on the upper unit body **80** instead of being provided on the upper support member **82**.

(49) As shown in FIGS. **1** and **6**, the bending system **10** includes a database **96** storing tool information about the plurality of die tools **14** and tool information about the plurality of punch tools **12**. The tool information about the plurality of die tools **14** is associated with identification information about the plurality of die tools **14** in the database **96**. The tool information about the plurality of punch tools **12** is associated with identification information about the plurality of punch tools **12** in the database **96**.

(50) The bending system **10** includes a control device **98** to control the hydraulic cylinders **28** and the like based on a processing program and controls the lower tool exchange units **50** and the like based on a tool exchange program. The control device **98** is configured with a computer, and the encoders **54**, the photoelectric sensors **68**, the two-dimensional code readers **70**, the encoders **78**, the photoelectric sensors **92**, the two-dimensional code readers **94**, the database **96** and the like are connected to the control device **98**. The control device **98** has a memory (not shown) storing the processing program, the tool exchange program and the like, and a CPU (not shown) that executes the processing program and the tool exchange program.

(51) The processing program and the tool exchange program include layout information showing a layout state of die tools **14** and layout information showing a layout state of die tools **14** on each lower stocker **44**. The processing program and the tool exchange program include layout information showing a layout state of the punch tools **12** and layout information showing a layout state of the punch tools **12** on each upper stocker **46**.

(52) By executing the tool exchange program, the CPU of the control device **98** demonstrates functions as a hole position acquisition unit **100**, a tool information acquisition unit **102**, a retry control unit **104** and a layout control unit **106**. Specific content of functions of the hole position acquisition unit **100**, the tool information acquisition unit **102**, the retry control unit **104** and the layout control unit **106** are as below.

(53) As shown in FIGS. **4** to **6**, based on positions of the lower tool exchange units **50** in the left-right direction at the time of engagement holes **14h** of die tools **14** being detected by the photoelectric sensors **68**, the hole position acquisition unit **100** acquires positions of centers of the engagement holes **14h** of the die tools **14** in the left-right direction by calculation. The hole position acquisition unit **100** acquires the positions of the centers of the engagement holes **14h** of the die tools **14** in the left-right direction in consideration of relative positions of the photoelectric sensors **68** relative to the lower unit bodies **56**. Similarly, based on positions of the upper tool exchange units **74** in the left-right direction at the time of engagement holes **12h** of punch tools **12** being detected by the photoelectric sensors **92**, the hole position acquisition unit **100** acquires positions of centers of the engagement holes **12h** of the punch tools **12** in the left-right direction by calculation. The hole position acquisition unit **100** acquires the positions of the centers of the engagement holes **12h** of the punch tools **12** in the left-right direction in consideration of a relative positions of the photoelectric sensors **92** relative to the upper unit bodies **80**.

(54) Referring to the database **96**, the tool information acquisition unit **102** acquires tool information about the die tools **14**, including width dimensions of the die tools **14** in the left-right direction and relative positions of the centers of the engagement holes **14h**, based on identification information about the die tools **14** from the two-dimensional code readers **70**. Similarly, referring to the database **96**, the tool information acquisition unit **102** acquires tool information about the punch tools **12**, including width dimensions of the punch tools **12** in the left-right direction and relative positions of the centers of the engagement holes **12h**, based on identification information about the punch tools **12** from the two-dimensional code readers **94**.

(55) As shown in FIGS. **3**, **4**, **6**, **7A** and **7B**, if the operation of retaining a die tool **14** by one of the lower tool retention members **64** fails (see **7(a)**), the retry control unit **104** executes a retry process of the lower tool exchange unit **50** and the like as below. Here, the failure in the operation of retaining the die tool **14** by the lower tool retention member **64** is judged by the fact that the air cylinder **60** or **66** has not reached a stroke end. If the operation of retaining the die tool **14** by the lower tool retention member **64** fails, an alarm occurs.

(56) The retry control unit **104** controls the servo motor **52** so that the lower tool exchange unit **50** moves in the left-right direction (for example, in the left direction) relative to the lower table **24** by an amount corresponding to a relative position of the engagement hole **14h** of the die tool **14**, which is tool information about the die tool **14** acquired immediately after the alarm occurs (see FIG. **7A(a)**). Next, the retry control unit **104** controls the servo motor **52** so that the lower tool exchange unit **50** moves in the left-right direction (for example, in the right direction) relative to the lower table **24** near a position where the alarm occurred (see FIGS. **7A(b)** and **7B(a)**). After that, based on a position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction acquired during movement of the lower tool exchange unit **50**, the retry control unit **104** controls the lower tool exchange unit **50** to retry the operation of retaining the die tool **14** by the lower tool retention member **64** (see FIG. **7B(b)**).

(57) If the operation of retaining a punch tool **12** by one of the upper tool retention members **88** fails, the retry control unit **104** also executes the retry process of the upper tool exchange unit **74** and the like similarly to the above.

(58) As shown in FIGS. **3**, **4**, **6**, **8A** and **8B**, if the clumper **32** is accidentally released after a plurality of die tools **14** are fitted to the lower tool holder **30**, the layout control unit **106** executes a layout process of the lower tool exchange units **50** and the like as below. Similarly, if a plurality of die tools **14** are randomly fitted to the lower tool holder **30** by manual work of a worker, the layout

control unit **106** also executes the layout process of the lower tool exchange units **50** and the like as necessary.

(59) The layout control unit **106** controls the servo motors **52** so that the lower tool exchange units **50** move in the left-right direction (for example, in the right direction) relative to the lower table **24** from the side of the lower table **24** (see FIGS. **8A(a)** and **8A(b)** and FIG. **8B(a)**). After that, based on acquired positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14**, the layout control unit **106** controls the lower tool exchange units **50** to lay out again the plurality of die tools **14** at regular arrangement positions on the lower tool holder **30** (see FIG. **8B(b)**). Alternatively, based on the positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14** on a lower stocker **44**, the layout control unit **106** controls the lower tool exchange units **50** to remove the plurality of die tools **14** from the lower tool holder **30** and lay out them at regular arrangement positions on the lower stocker **44** as necessary.

(60) The operation of laying out die tools **14** by the lower tool exchange units **50** is commonly known as shown in Patent Literatures 2 and 3, details thereof will be omitted.

(61) If the clamp plate **36** is accidentally released after a plurality of punch tools **12** are fitted to the upper tool holder **34**, the layout control unit **106** also executes a layout process of the upper tool exchange units **74** and the like similarly to the above.

(62) Next, operation and effects of the first embodiment will be explained.

(63) In the case of transporting die tools **14** from the lower tool holder **30** to a lower stocker **44** positioned at the lower tool exchange position, the control device **98** controls the servo motor **52** to cause the lower tool exchange units **50** to move in the left-right direction and causes the lower tool retention members **64** to face the engagement holes **14h** of the die tools **14** fitted to the lower tool holder **30**. Then, while causing the lower tool retention members **64** to be engaged with the engagement holes **14h** of the die tools **14**, the control device **98** causes the lower tool retention members **64** to retain the die tools **14**. Furthermore, the control device **98** controls the lower tool exchange units **50** to remove the die tools **14** from the lower tool holder **30**. The die tools **14** may be removed from the lower tool holder **30** by being caused to slide in the right direction on the lower tool holder **30**.

(64) Next, the control device **98** controls the servo motors **52** to cause the lower tool exchange units **50** to move in the right direction and cause the lower tool retention members **64** to be located at positions corresponding to the lower stocker **44** positioned at the lower tool exchange position. Then, the control device **98** controls the lower tool exchange units **50** to fit the die tools **14** to the lower stocker **44**. After that, the control device **98** causes the lower tool retention members **64** to be disengaged from the engagement holes **14h** of the die tools **14** and releases the state of retaining the die tools **14** by the lower tool retention members **64**. The die tool **14** may be fitted to the lower stocker **44** by being caused to slide in the right direction on the lower joint block **38**.

(65) In the case of transporting die tools **14** from a lower stocker **44** positioned at the lower tool exchange position to the lower tool holder **30**, an operation opposite to the operation described above is performed. Thereby, it is possible to perform automatic exchange of die tools **14** fitted to the lower tool holder **30** with die tools **14** fitted to the lower stocker **44**.

(66) In the case of transporting punch tools **12** between the upper tool holder **34** and an upper stocker **46** positioned at the upper tool exchange position, an operation similar to the above is also performed. Thereby, it is possible to perform automatic exchange of punch tools **12** fitted to the upper tool holder **34** with punch tools **12** fitted to an upper stocker **46**.

(67) If the operation of retaining a die tool **14** by one of the lower tool retention members **64** fails in the case of transporting die tools **14** from the lower tool holder **30** to a lower stocker **44** positioned at the lower tool exchange position (see FIG. **7A(a)**), the following operation is performed.

(68) Immediately after an alarm occurs, while referring to the database **96**, the tool information acquisition unit **102** acquires tool information about the die tool **14** including a relative position of the engagement hole **14h** of the die tool **14**, based on identification information about the die tool **14** from the two-dimensional code reader **70**. Then, the retry control unit **104** controls the servo motor **52** to cause the lower tool exchange unit **50** to move in the left-right direction (for example, in the left direction) relative to the lower table **24** by an amount corresponding to the relative position of the engagement hole **14h** of the die tool **14** (see FIG. 7A(a)). The tool information acquisition unit **102** may acquire tool information about the die tool **14** based on a detection result from the encoder **54** at the time of the occurrence of the alarm and layout information about the die tool **14**.

(69) Next, the retry control unit **104** controls the servo motor **52** to cause the lower tool exchange unit **50** to move in the left-right direction near the alarm occurrence position while monitoring detection results from the encoder **54** (see FIG. 7A(b)). Further, during the movement of the lower tool exchange unit **50**, the photoelectric sensor **68** emits an inspection beam B and detects the engagement hole **14h** of the die tool **14** from a result of receiving a reflected beam of the inspection beam B (see FIG. 7B(a)). Then, based on a position of the lower tool exchange unit **50** in the left-right direction at the time of the engagement hole **14h** of the die tool **14** being detected and the like, the hole position acquisition unit **100** acquires a position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction.

(70) After that, based on the acquired position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction, the retry control unit **104** controls the lower tool exchange unit **50** to retry the operation of retaining the die tool **14** by the lower tool retention member **64** (see FIG. 7B (b)). Thereby, it is possible to continue the operation related to the transportation of the die tools **14** by the lower tool exchange units **50**.

(71) If the operation of retaining a punch tool **12** by one of the upper tool retention members **88** fails in the case of transporting punch tools **12** from the upper tool holder **34** to an upper stocker **46** positioned at the upper tool exchange position, an operation similar to the above is also performed. Thereby, it is possible to continue the operation related to the transportation of the punch tools **12** by the upper tool exchange units **74**.

(72) In a case where the clamper **32** is accidentally released after a plurality of die tools **14** are fitted to the lower tool holder **30**, a case where a plurality of die tools **14** are randomly fitted to the lower tool holder **30**, or the like, the following operation is performed.

(73) The layout control unit **106** controls the servo motors **52** to cause the lower tool exchange units **50** to move in the left-right direction (for example, in the right direction) relative to the lower table **24** from the side of the lower table **24** while monitoring detection results from the encoders **54** (see FIGS. 8A(a) and 8A(b) and FIG. 8B(a)). During the movements of the lower tool exchange units **50**, the photoelectric sensors **68** emit inspection beams B and detect the engagement holes **14h** of the plurality of die tools **14** from results of receiving reflected beams of the inspection beams B. Then, based on positions of the lower tool exchange units **50** in the left-right direction at the time of the engagement holes **14h** of the plurality of die tools **14** being detected and the like, the hole position acquisition unit **100** acquires positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction (see FIG. 8A(b)). Thereby, the bending system **10** can recognize an actual layout state of the plurality of die tools **14** on the lower tool holder **30**. During the movements of the lower tool exchange units **50**, the tool information acquisition unit **102** may acquire tool information about the plurality of die tools **14**.

(74) After that, based on the acquired positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14**, the layout control unit **106** controls the lower tool exchange units **50** to lay out again the plurality of die tools **14** at regular arrangement positions on the lower tool holder **30** (see FIG. 8B(b)). Alternatively, based on the acquired positions of the centers of the engagement holes **14h** of

the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14** on a lower stocker **44**, the layout control unit **106** controls the lower tool exchange units **50** to remove the plurality of die tools **14** from the lower tool holder **30** and lay out them at regular arrangement positions on the lower stocker **44**. Thereby, it is possible to continue the operation related to the transportation of the die tools **14** by the lower tool exchange units **50**.

(75) In a case where the clamp plate **36** is accidentally released after a plurality of punch tools **12** are fitted to the upper tool holder **34**, a case where a plurality of punch tools **12** are randomly fitted to the upper tool holder **34**, or the like, an operation similar to the above is also performed.

Thereby, it is possible to continue the operation related to the transportation of the punch tools **12** by the upper tool exchange units **74**.

(76) If there are gaps among a plurality of die tools **14** during movement of the lower tool exchange units **50**, the photoelectric sensors **68** detect the end faces **14e** of die tools **14** on the middle side.

The bending system **10** can determine whether there are gaps among the plurality of die tools **14**, based on tool layout information about the plurality of die tools **14**.

(77) In short, according to the first embodiment, based on positions of the lower tool exchange units **50** in the left-right direction at the time of the engagement holes **14h** of die tools **14** being detected by the photoelectric sensors **68**, and the like, the hole position acquisition unit **100** acquires positions of the centers of the engagement holes **14h** of the die tools **14** in the left-right direction by calculation, as mentioned above. Based on positions of the upper tool exchange units **74** in the left-right direction at the time of the engagement holes **12h** of punch tools **12** being detected by the photoelectric sensors **92** and the like, the hole position acquisition unit **100** acquires positions of the centers of the engagement holes **12h** of the punch tools **12** in the left-right direction. Further, the tool information acquisition unit **102** acquires tool information about die tools **14** based on identification information about the die tools **14** from the two-dimensional code readers **70**. The tool information acquisition unit **102** acquires tool information about punch tools **12** based on identification information about the punch tools **12** from the two-dimensional code readers **94**. Thereby, the bending system **10** can accurately recognize actual positions of the engagement holes **14h** of a plurality of die tools **14** in the left-right direction on the lower tool holder **30** and actual positions of the engagement holes **12h** of a plurality of punch tools **12** in the left-right direction on the upper tool holder **34**.

(78) Therefore, according to the first embodiment, it is possible to stabilize automatic operation of the bending system **10** without the operation related to transportation of die tools **14** by the lower tool exchange units **50** and the like being interrupted in the middle of the operation and the like.

Second Embodiment

(79) As shown in FIG. **1**, a bending system **108** according to a second embodiment has a configuration similar to that of the bending system **10** according to the first embodiment except for a part thereof. Among components of the bending system **108**, components different from those of the bending system **10** will be explained. Among the plurality of components of the bending system **108**, components corresponding to components of the bending system **10** are given the same reference signs in drawings.

(80) As shown in FIGS. **6** and **9**, at an appropriate position on each lower support member **58**, a photoelectric sensor **110** is provided as an end face detector to detect the end face **14e** of a die tool **14** fitted to the lower tool holder **30** instead of the photoelectric sensor **68** (see FIG. **4**). The photoelectric sensor **110** is configured to emit an inspection beam B and detect the end face **14e** of the die tool **14** from a result of receiving a reflected beam of the inspection beam B, in a state of being located at the same height as a part higher than the engagement hole **14h** on the die tool **14**.

(81) The photoelectric sensor **110** may detect the end face **14e** of the die tool **14** in a state of being located at the same height as the engagement hole **14h** of the die tool **14**. In this case, the control device **98** judges which of the end face **14e** and engagement hole **14h** of the die tool **14** the detection target of the photoelectric sensor **110** is, based on tool layout information and the like

about a plurality of die tools **14**.

(82) As shown in FIG. **10**, at an appropriate position on each upper support member **82**, a photoelectric sensor **112** is provided as an end face detector to detect the end face **12e** of a punch tool **12** fitted to the upper tool holder **34** instead of the photoelectric sensor **92** (see FIG. **5**). The photoelectric sensor **112** is configured to emit an inspection beam B and detect the end face **12e** of the punch tool **12** from a result of receiving a reflected beam of the inspection beam B, in a state of being located at the same height as a part lower than the engagement hole **12h** on the punch tool **12**.

(83) The photoelectric sensor **112** may detect the end face **12e** of the punch tool **12** in a state of being located at the same height as the engagement hole **12h** of the punch tool **12**. In this case, the control device **98** judges which of the end face **12e** and engagement hole **12h** of the punch tool **12** the detection target of the photoelectric sensor **110** is, based on tool layout information and the like about a plurality of punch tools **12**.

(84) As shown in FIG. **6**, in the second embodiment, the control device **98** has a hole position acquisition unit **114**, a retry control unit **116** and a layout control unit **118** instead of the hole position acquisition unit **100**, the retry control unit **104** and the layout control unit **106**. Specific content of functions of the hole position acquisition unit **114**, the retry control unit **116** and the layout control unit **118** are as below.

(85) As shown in FIGS. **6**, **9** and **10**, based on positions of the lower tool exchange units **50** in the left-right direction at the time of end faces **14e** of die tools **14** being detected, and acquired tool information about the die tools **14** (relative positions of the engagement holes **14h** of the die tools **14**), the hole position acquisition unit **114** acquires positions of the centers of the engagement holes **14h** of the die tools **14** in the left-right direction by calculation. The hole position acquisition unit **114** acquires the positions of the centers of the engagement holes **14h** of the die tools **14** in the left-right direction in consideration of relative positions of the photoelectric sensors **110** relative to the lower unit bodies **56**. Similarly, based on positions of the upper tool exchange units **74** in the left-right direction at the time of end faces **12e** of punch tools **12** being detected, and acquired tool information about the punch tools **12** (relative positions of the engagement holes **12h** of the punch tools **12**), the hole position acquisition unit **114** acquires positions of the centers of the engagement holes **12h** of the punch tools **12** in the left-right direction by calculation. The hole position acquisition unit **114** acquires the positions of the centers of the engagement holes **12h** of the punch tools **12** in the left-right direction in consideration of relative positions of the photoelectric sensors **112** relative to the upper unit bodies **80**.

(86) As shown in FIGS. **3**, **6**, **9**, **11A** and **11B**, if the operation of retaining a die tool **14** by one of the lower tool retention members **64** fails (see **11A(a)**), the retry control unit **116** executes a retry process of the lower tool exchange unit **50** and the like as below.

(87) The retry control unit **116** controls the servo motor **52** so that the lower tool exchange unit **50** moves in the left-right direction (for example, in the left direction) relative to the lower table **24** by an amount corresponding to a relative position of the engagement hole **14h** of the die tool **14**, which is tool information about the die tool **14** acquired immediately after an alarm occurs (see FIG. **11A(a)**). Next, the retry control unit **116** controls the servo motor **52** so that the lower tool exchange unit **50** moves in the left-right direction (for example, in the right direction) relative to the lower table **24** from the side of the die tool **14** (see FIGS. **11A(b)** and **11B(a)**). After that, based on an acquired position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction, the retry control unit **116** controls the lower tool exchange unit **50** to retry the operation of retaining the die tool **14** by the lower tool retention member **64** (see FIG. **11B(b)**).

(88) If the operation of retaining a punch tool **12** by one of the upper tool retention members **88** fails, the retry control unit **116** also executes the retry process of the upper tool exchange unit **74** and the like similarly to the above.

(89) As shown in FIGS. **3**, **6**, **9**, **12A** and **12B**, if the clamber **32** is accidentally released after a plurality of die tools **14** are fitted to the lower tool holder **30**, the layout control unit **118** executes a

layout process of the lower tool exchange units **50** and the like as below. Similarly, if a plurality of die tools **14** are randomly fitted to the lower tool holder **30** by manual work of a worker, the layout control unit **118** also executes the layout process of the lower tool exchange units **50** and the like as necessary.

(90) The layout control unit **118** controls the servo motors **52** so that the lower tool exchange units **50** move in the left-right direction (for example, in the right direction) relative to the lower table **24** from the side of the lower table **24** (see FIGS. **12A(a)** and **12A(b)** and FIG. **12B(a)**). After that, based on acquired positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14**, the layout control unit **118** controls the lower tool exchange units **50** to lay out again the plurality of die tools **14** at regular arrangement positions on the lower tool holder **30** (see FIG. **12B(b)**).

Alternatively, based on the positions of centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14** on a lower stocker **44**, the layout control unit **118** controls the lower tool exchange units **50** to remove the plurality of die tools **14** from the lower tool holder **30** and lay out them at regular arrangement positions on the lower stocker **44** as necessary.

(91) If the clamp plate **36** is accidentally released after a plurality of punch tools **12** are fitted to the upper tool holder **34**, the layout control unit **118** also executes a layout process of the upper tool exchange units **74** and the like similarly to the above.

(92) Next, operation and effects of the second embodiment will be explained.

(93) If the operation of retaining a die tool **14** by one of the lower tool retention members **64** fails in the case of transporting the die tools **14** from the lower tool holder **30** to a lower stocker **44** positioned at the lower tool exchange position (see FIG. **11A(a)**), the following operation is performed.

(94) Immediately after an alarm occurs, while referring to the database **96**, the tool information acquisition unit **102** acquires tool information about the die tool **14** including a relative position of the engagement hole **14h** of the die tool **14**, based on identification information about the die tool **14** from the two-dimensional code reader **70**. Then, the retry control unit **116** controls the servo motor **52** to cause the lower tool exchange unit **50** to move in the left-right direction (for example, in the left direction) relative to the lower table **24** by an amount corresponding to the relative position of the engagement hole **14h** of the die tool **14** (see FIG. **11A(a)**). The tool information acquisition unit **102** may acquire tool information about the die tool **14** based on a detection result from the encoder **54** at the time of the occurrence of the alarm and layout information about the die tool **14**.

(95) Next, the retry control unit **116** controls the servo motor **52** to cause the lower tool exchange unit **50** to move in the left-right direction (for example, in the right direction) from the side of the die tool **14** while monitoring detection results from the encoder **54** (see FIG. **11A(b)**). Further, during the movement of the lower tool exchange unit **50**, the photoelectric sensor **110** emits an inspection beam B and detects the end face **14e** of the die tool **14** from a result of receiving a reflected beam of the inspection beam B (see FIG. **11B(a)**). Then, based on a position of the lower tool exchange unit **50** in the left-right direction at the time of the end face **14e** of the die tool **14** being detected, acquired tool information about the die tool **14** and the like, the hole position acquisition unit **114** acquires a position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction.

(96) After that, based on the acquired position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction, the retry control unit **116** controls the lower tool exchange unit **50** to retry the operation of retaining the die tool **14** by the lower tool retention member **64** (see FIG. **11B(b)**). Thereby, it is possible to continue the operation related to the transportation of the die tools **14** by the lower tool exchange units **50**.

(97) If the operation of retaining a punch tool **12** by one of the upper tool retention members **88**

fails in the case of transporting punch tools **12** from the upper tool holder **34** to an upper stocker **46** positioned at the upper tool exchange position, an operation similar to the above is also performed. Thereby, it is possible to continue the operation related to the transportation of the punch tools **12** by the upper tool exchange units **74**.

(98) In a case where the clamper **32** is accidentally released after a plurality of die tools **14** are fitted to the lower tool holder **30**, a case where a plurality of die tools **14** are randomly fitted to the lower tool holder **30**, or the like, the following operation is performed.

(99) The layout control unit **118** controls the servo motors **52** to cause the lower tool exchange units **50** to move in the left-right direction (for example, in the right direction) relative to the lower table **24** from the side of the lower table **24** (see FIGS. **12A(a)** and **12A(b)** and FIG. **12B(a)**). During the movements of the lower tool exchange units **50**, the photoelectric sensors **110** emit inspection beams B and detect the end face **14e** of a die tool **14** on the left end side from results of receiving reflected beams of the inspection beams B. The tool information acquisition unit **102** acquires tool information about the plurality of die tools **14** (width dimensions of the die tools **14** in the left-right direction and relative positions of the centers of the engagement holes **14h**). Further, based on positions of the lower tool exchange units **50** in the left-right direction at the time of the end face **14e** of the die tool **14** on the left end side being detected, the acquired tool information about the plurality of die tools **14** and the like, the hole position acquisition unit **114** acquires positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction (see FIG. **12A(b)**). Thereby, the bending system **10** can recognize an actual layout state of the plurality of die tools **14** on the lower tool holder **30**.

(100) After that, based on the acquired positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14**, the layout control unit **118** controls the lower tool exchange units **50** to lay out again the plurality of die tools **14** at regular arrangement positions on the lower tool holder **30** (see FIG. **12B(b)**). Alternatively, based on the acquired positions of the centers of the engagement holes **14h** of the plurality of die tools **14** in the left-right direction and tool layout information about the plurality of die tools **14** on a lower stocker **44**, the layout control unit **118** controls the lower tool exchange units **50** to remove the plurality of die tools **14** from the lower tool holder **30** and lay out them at regular arrangement positions on the lower stocker **44**. Thereby, it is possible to continue the operation related to the transportation of the die tools **14** by the lower tool exchange units **50**.

(101) If there are gaps among the plurality of die tools **14** during the movements of the lower tool exchange units **50**, the photoelectric sensors **110** detect the end faces **14e** of die tool **14** on the middle side. The bending system **10** can determine whether there are gaps among the plurality of die tools **14**, based on tool layout information about the plurality of die tools **14**.

(102) In a case where the clamp plate **36** is accidentally released after a plurality of punch tools **12** are fitted to the upper tool holder **34**, a case where a plurality of punch tools **12** are randomly fitted to the upper tool holder **34**, or the like, an operation similar to the above is also performed. Thereby, it is possible to continue the operation related to the transportation of the punch tools **12** by the upper tool exchange units **74**.

(103) In short, according to the second embodiment, based on positions of the lower tool exchange units **50** in the left-right direction at the time of the end faces **14e** of die tools **14** being detected by the photoelectric sensors **110**, and the like, the hole position acquisition unit **114** acquires positions of the centers of the engagement holes **14h** of the die tools **14** in the left-right direction by calculation, as mentioned above. Based on positions of the upper tool exchange units **74** in the left-right direction at the time of the end faces **12e** of punch tools **12** being detected by the photoelectric sensors **112** and the like, the hole position acquisition unit **114** acquires positions of the centers of the engagement holes **12h** of the punch tools **12** in the left-right direction. Further, the tool information acquisition unit **102** acquires tool information about die tools **14** based on identification information about the die tools **14** from the two-dimensional code readers **70**. The

tool information acquisition unit **102** acquires tool information about punch tools **12** based on identification information about the punch tools **12** from the two-dimensional code readers **94**. Thereby, the bending system **10** can accurately recognize actual positions of the engagement holes **14h** of a plurality of die tools **14** in the left-right direction on the lower tool holder **30** and actual positions of the engagement holes **12h** of a plurality of punch tools **12** in the left-right direction on the upper tool holder **34**.

(104) Therefore, according to the second embodiment, it is possible to stabilize automatic operation of the bending system **10** without the operation related to transportation of die tools **14** by the lower tool exchange units **50** and the like being interrupted in the middle of the operation and the like.

Third Embodiment

(105) As shown in FIG. **13**, a bending system **120** according to a third embodiment uses torque sensors **122** to detect torques of the servo motors **52** as end face detectors to detect the end faces **14e** of die tools **14**, instead of the photoelectric sensors **110** (see FIG. **6**). Each torque sensors **122** is configured to detect the end face **14e** of a die tool **14** when a detected torque of its corresponding servo motor **52** (a detection result) exceeds a predetermined threshold. Similarly, the bending system **120** uses torque sensors **124** to detect torques of servo motors **75** as end face detectors to detect the end faces **12e** of punch tools **12**, instead of the photoelectric sensors **112** (see FIG. **6**). Each torque sensor **124** is configured to detect the end face **12e** of a punch tool **12** when a detected torque of the servo motor **75** exceeds a predetermined threshold.

(106) In the third embodiment, the control device **98** has an end face detection control unit **126** and has a hole position acquisition unit **128** instead of the hole position acquisition unit **100** (see FIG. **6**). Specific content of functions of the end face detection control unit **126** is as below.

(107) As shown in FIGS. **13** and **14**, before detecting the end faces **14e** of die tools **14**, the end face detection control unit **126** controls one lower tool exchange unit **50** so that one lower tool retention member **64** faces an end face **14e** on one side of the die tools **14** (for example, an end face on the left side) in the left-right direction (see FIG. **14(a)**). Further, before detecting the end face **14e** on the one side of the die tools **14**, the end face detection control unit **126** controls the other lower tool exchange unit **50** so that the other lower tool retention member **64** retains an engagement hole **14h** on the other side of the die tools **14**, or the retention state is continued (see FIGS. **14(a)** and **(b)**). Alternatively, before detecting the end face **14e** on the one side of the die tools **14**, the end face detection control unit **126** controls the other lower tool exchange unit **50** so that the other lower tool retention member **64** is in contact with or close to the end face **14e** on the other side (for example, an end face on the right side) of the die tools **14**.

(108) Similarly, before detecting the end faces **12e** of punch tools **12**, the end face detection control unit **126** controls one upper tool exchange unit **74** so that one upper tool retention member **88** faces an end face **12e** on one side of the punch tools **12** (for example, an end face on the left side) in the left-right direction. Further, before detecting the end face **12e** on the one side of the punch tools **12**, the end face detection control unit **126** controls the other upper tool exchange unit **74** so that the other upper tool retention member **88** retains the engagement hole **12h** of a punch tool **12**, or the retention state is maintained. Alternatively, before detecting the end face **12e** on the one side of the punch tools **12**, the end face detection control unit **126** controls the other upper tool exchange unit **74** so that the other upper tool retention member **88** is in contact with or close to an end face **12e** on the other side (for example, an end face on the right side) of the punch tools **12**.

(109) Based on a position of the one lower tool exchange unit **50** in the left-right direction at the time of the end face **14e** on the one side of the die tools **14** being detected, and acquired tool information about the die tool **14** (a relative position of the engagement hole **14h** of the die tool **14**), the hole position acquisition unit **128** acquires a position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction by calculation. The hole position acquisition unit **128** acquires the position of the center of the engagement hole **14h** of the die tool **14** in the left-right direction in consideration of a relative position of the one lower tool retention member **64**

relative to the lower unit body **56**. Similarly, based on a position of the one upper tool exchange unit **74** in the left-right direction at the time of the end face **12e** on the one side of the punch tools **12** being detected, and acquired tool information about the punch tool **12** (a relative position of the engagement hole **12h** of the punch tool **12**), the hole position acquisition unit **128** acquires a position of the center of the engagement hole **12h** of the punch tool **12** in the left-right direction by calculation. The hole position acquisition unit **128** acquires the position of the center of the engagement hole **12h** of the punch tool **12** in the left-right direction in consideration of a relative position of the one upper tool retention member **88** relative to the upper unit body **80**.

(110) The bending system **120** performs an operation related to detection of the end faces **14e** of the die tools **14** as below.

(111) The end face detection control unit **126** controls one lower tool exchange unit **50** so that one lower tool retention member **64** faces one end face **14e** of a die tool **14** on the left end side in the left-right direction. Further, the end face detection control unit **126** controls the other lower tool exchange unit **50** so that the other lower tool retention member **64** retains the engagement hole **14h** of a die tool **14** on the right end side, or the retention state is continued (see FIG. **14(a)**). Alternatively, the end face detection control unit **126** controls the other lower tool exchange unit **50** so that the other lower tool retention member **64** is in contact with or close to the other end face **14e** of the die tool **14** on the right end side. Then, the retry control unit **116** or a layout control unit **119** controls the servo motors **52** to cause the lower tool exchange units **50** to move in the left-right direction (for example, in the right direction) from the side of the die tools **14** while monitoring detection results from the encoders **54**.

(112) Then, the one lower tool retention member **64** comes into contact with the one end face **14e** of the die tool **14** on the left end side; a detection torque of one servo motor **52** exceeds the predetermined threshold; and one torque sensor **122** detects the one end face **14e** of the die tool **14** on the left end side (see FIG. **14(b)**). Based on a position of the one lower tool exchange unit **50** in the left-right direction at the time of the one end face **14e** of the die tool **14** on the left end side being detected, the hole position acquisition unit **114** acquires a position of the center of the engagement hole **14h** of the die tool **14** on the left end side in the left-right direction by calculation. Further, in a state of the other lower tool retention member **64** being in contact with the other end face **14e** of the die tool **14** on the right end side, a detection torque of the other servo motor **52** exceeds the predetermined threshold, and the other torque sensor **122** detects the other end face **14e** of the die tool **14** on the right end side. Based on a position of the other lower tool exchange unit **50** in the left-right direction at the time of the other end face **14e** of the die tool **14** on the right end side being detected and the like, the hole position acquisition unit **114** acquires a position of the center of the engagement hole **14h** of the die tool **14** on the right end side in the left-right direction by calculation.

(113) The bending system **120** performs an operation related to detection of the end face **12e** of a punch tool **12** similarly to the above.

(114) According to the third embodiment, operation and effects similar to those of the second embodiment described above can be obtained.

Fourth Embodiment

(115) As shown in FIG. **15**, a part of the technical items described in the second embodiment may be applied to a general-purpose bending system **130**. The bending system **130** according to a fourth embodiment does not include the tool exchange units **50** or **74** (see FIG. **6**), the tool rack **42** (see FIG. **6**), the servo motors **52** or **76** (see FIG. **6**) or the like. The photoelectric sensors **110** and the two-dimensional code readers **70** are provided at appropriate positions on sliders **132** with back gauge mechanisms as moving bodies, respectively. Servo motors **134** as left-right moving actuators to cause the sliders **132** to move in the left-right direction are connected to the control device **98** of the bending system **130**. Encoders **136** as position detectors to detect positions of the sliders **132** in the left-right direction are connected to the control device **98** of the bending system **130**.

(116) The control device **98** of the bending system **120** includes a tool position acquisition unit **138** instead of the hole position acquisition unit **114** (see FIG. 6). Based on positions of the sliders **132** in the left-right direction at the time of the end faces **14e** (see FIG. 9) of die tools **14** being detected, and acquired tool information (width dimensions of the die tools **14** in the left-right direction) about the die tools **14** (see FIG. 1), the tool position acquisition unit **138** acquires positions of the end faces **14e** of the die tools **14** in the left-right direction by calculation. Similarly, based on positions of the other slider **132** in the left-right direction at the time of the end faces **12e** (see FIG. 9) of punch tools **12** being detected, and acquired tool information (width dimensions of the punch tools **12** in the left-right direction) about the punch tools **12**, the tool position acquisition unit **138** acquires positions of the end faces **12e** of the punch tools **12** in the left-right direction by calculation. Thereby, the bending system **130** can accurately recognize actual positions of end faces **14e** of a plurality of die tools **14** in the left-right direction on the lower tool holder **30** (see FIG. 1) and actual positions of end faces **12e** of a plurality of punch tools **12** in the left-right direction on the upper tool holder **34** (see FIG. 1).

(117) In the fourth embodiment, a die tool **14** does not need to have an engagement hole **14h**, and a punch tool **12** does not need to have an engagement hole **12h**.

(118) The present disclosure is not limited to the explanations of the first to fourth embodiment described above. For example, the present disclosure can be implemented in various aspects as below.

(119) As hole detectors to detect the engagement holes **14h** of die tools **14**, cameras to pick up images of the engagement holes **14h** of the die tools **14** may be used instead of using the photoelectric sensors **68**. In this case, using a reference image for the engagement holes **14h** of the die tools **14**, the hole position acquisition unit **100** judges whether a picked-up image corresponds to the reference image or not by pattern matching. Similarly, as hole detectors to detect the engagement hole **12h** of punch tools **12**, cameras to pick up images of the engagement holes **12h** of the punch tools **12** may be used instead of using the photoelectric sensors **92**.

(120) Instead of providing the identification marks **14m** to a die tool **14**, an IC chip storing tool information about the die tool **14** may be attached. Similarly, instead of providing the identification marks **12m** to a punch tool **12**, an IC chip storing tool information about the punch tool **12** may be attached. In this case, an IC chip reader as an information reader reads the tool information about the die tool **14** and the tool information about the punch tool **12**.

(121) The photoelectric sensors **68** (or **108**) and the two-dimensional code readers **70** may be provided on sliders **132** (see FIG. 15) with back gauge mechanisms as moving bodies provided on the back side of the lower table **24** being movable in the left-right direction, instead of being provided on the lower tool exchange units **50**. Alternatively, the photoelectric sensors **68** (or **108**) and the two-dimensional code readers **70** may be provided on bending indicators (not shown) as moving bodies provided on the front side of the lower table **24** being movable in the left-right direction.

(122) The operation related to tool exchange may be performed by a tool change robot (not shown) instead of performing the operation related to tool exchange by the lower tool exchange units **50** and the upper tool exchange units **74**.

(123) The scope of rights included in the present invention is not limited to the first to fourth embodiments described above.

(124) The present disclosure is related to the subjects described in Japanese Patent Application 2019-107126 filed on Jun. 7, 2019 and Japanese Patent Application 2020-093079 filed on May 28, 2020, the disclosed content of which is incorporated herein by reference in its entirety.

Claims

1. A bending system comprising: a press brake including a table and a tool holder provided on the table, a plurality of tools being fitted to the tool holder; a tool changer provided on a back side or a front side of a table of the press brake so as to be movable in a left-right direction via a guide extending in the left-right direction; a moving actuator configured to cause the tool changer to move in the left-right direction relative to the table; a position detector configured to detect a position of the tool changer in the left-right direction; a hole detector configured to detect an engagement hole for causing a hook of the tool changer to be engaged and disengaged, the engagement hole being a part of a tool of the plurality of tools fitted to the tool holder provided on the table; and a control device including a memory and a CPU, wherein the CPU is configured to acquire a position of the engagement hole of the tool in the left-right direction, based on the position of the tool changer in the left-right direction at the time of the engagement hole of the tool being detected, wherein the hole detector is a photoelectric sensor that is provided on the tool changer, and the hole detector and the engagement hole of the tool are located at a same height as one another when the tool is fitted to the tool holder provided on the table, and the hole detector is configured, during a movement of the tool changer in the left-right direction, to emit an inspection beam to the engagement hole of the tool and to detect the engagement hole of the tool from a result of receiving a reflected beam of the inspection beam.
 2. The bending system according to claim 1, further comprising: an information reader provided on the tool changer, the information reader being configured to read, from a storage medium provided to the tool, tool information including a width dimension of the tool in the left-right direction and a relative position of the engagement hole in the tool.
 3. The bending system according to claim 1, wherein the CPU is configured to, if an operation of retaining the tool by the hook fails, control the moving actuator so that the tool changer moves in the left-right direction relative to the table and to control the tool changer to retry the operation of retaining the tool by the hook based on the position of the engagement hole of the tool in the left-right direction acquired during the movement of the tool changer.
 4. The bending system according to claim 1, wherein the CPU is configured to control the moving actuator so that the tool changer moves in the left-right direction relative to the table and, based on the position of the engagement hole of the tool in the left-right direction acquired during the movement of the tool changer, and tool layout information about the tool, to control the tool changer to position the tool on the tool holder.
 5. The bending system according to claim 1, wherein the CPU is configured to control the moving actuator so that the tool changer moves in the left-right direction relative to the table and, based on the position of the engagement hole of the tool in the left-right direction acquired during the movement of the tool changer, and tool layout information about the tool on a stocker in a tool rack, to control the tool changer to cause the tool to be removed from the tool holder and store the tool on the stocker.
 6. The bending system according to claim 1, wherein the tool changer is configured to transport the tool between the tool holder and a stocker in a tool rack.
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